

PUV Nearshore Wave Measurement Pipeline

Processing, Validation, and Spectral Bias Investigation

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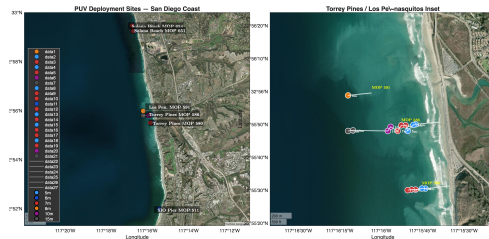
Outline

- 1 Overview
- 2 L1 Processing
- 3 L2 Spectral Analysis
- 4 Validation Against MOP
- 5 Hypothesis Testing
- 6 L3 Forcing Characterization
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- 9 Key Findings and Next Steps

- Nearshore wave transformation governs beach morphology, sediment transport, and coastal hazards
- Accurate in-situ wave measurements at 5–15 m depth are needed to validate spectral wave models and drive transport calculations
- PUV (pressure–velocity) measurements from bottom-mounted ADVs provide directional spectra without surface moorings
- Goal: build a robust, reproducible processing pipeline from raw Nortek Vector data to validated wave statistics

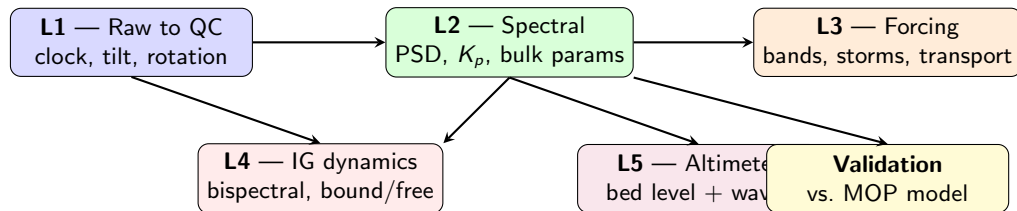
Instrument Network

- **20 deployments, 40 instruments**
- 5 sites along San Diego coast:
 - Torrey Pines — MOP 580 & 586
 - SIO Pier — MOP 511
 - Solana Beach — MOP 654
 - Los Peñasquitos Lagoon — MOP 591
- Depths: 5–15 m
- Deployment period: 2023–2026
- Nortek Vector ADVs, 2 Hz continuous



Five sites along the San Diego coast with MOP transect lines. Symbols mark PUV instrument locations at 5–15 m depth.

Pipeline Architecture



- L1–L3: **universal** — same outputs for any deployment
- L4–L5: **analysis-specific** — IG dynamics and bed-level integration
- All code: MATLAB, GitHub-tracked

GitHub: [holdenlesliebole/Nortek_Vector_PUV_Pipeline](https://github.com/holdenlesliebole/Nortek_Vector_PUV_Pipeline)

Raw Data Ingestion

- Nortek Vector burst files: `.dat` (data), `.sen` (sensor/diagnostics), `.hdr` (header)
- Parsed with MATLAB `textscan` for speed — deployments can exceed 10^8 samples
- Continuous 2 Hz time series reconstructed from burst boundaries
- **Clock drift correction:** linear interpolation between deployment and recovery clocks
- Battery cutoff detection: $\text{gaps} > 1 \text{ s} \Rightarrow$ discard remainder of record

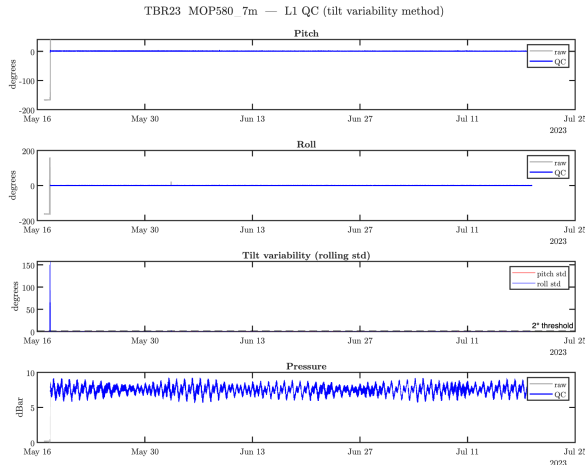
Quality Control — Old vs. New Approach

Old: Fixed threshold

- Absolute $\pm 5^\circ$ pitch/roll cap
- Discards all tilted data
- Problematic when pipes bend during storms — stable but non-zero tilt is *correctable*

New: Variability-based QC

- Rolling std of pitch/roll over 60 s windows
- Flag samples where variability $> 2^\circ$
- Absolute cap retained at 30° for severely compromised instruments
- Stable tilts *retained* and corrected



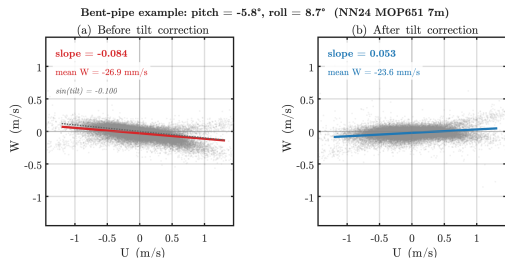
L1 QC diagnostic for TBR23 MOP580 at 8.3 m depth. Top two panels: pitch and roll time series (blue) with QC flags (red). Third panel: rolling standard deviation of tilt with the 2° variability

Tilt Correction

Sample-by-sample 3D rotation using measured pitch (ϕ) and roll (θ):

$$\mathbf{R} = \mathbf{R}_{\text{roll}}(\theta) \mathbf{R}_{\text{pitch}}(\phi)$$
$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \phi & 0 & \sin \phi \\ 0 & 1 & 0 \\ -\sin \phi & 0 & \cos \phi \end{bmatrix}$$

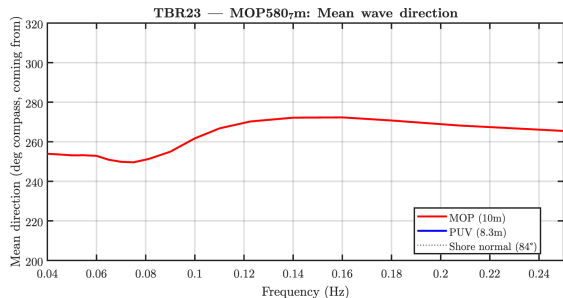
- Applied *before* heading rotation to buoy coordinates
- Corrects velocity components for instrument tilt
- Recovers data from bent mounting pipes (post-storm)



U-W scatter for SOL23 MOP651 (7 m) with pitch = -5.8° , roll = 8.7° (bent pipe). (a) Before correction: slope = -0.084 , matching $\sin(\text{tilt}) = -0.100$ (dotted). (b) After rotation: slope reduced to $+0.053$.

Coordinate Rotation

- Instrument XYZ $\rightarrow$ geographic buoy coordinates
- Convention: $+x$ West, $+y$ North, $+z$ Up (left-handed, MOP-consistent)
- Heading from deployment notes + magnetic declination via IGRF-13 model at deployment location and epoch
- Shore-normal rotation applied later in L2 using CDIP THREDDS shore-normal angle for each MOP station



Mean wave direction vs. frequency for TBR23 MOP580 at 8.3 m.
Red: MOP model (10 m). Blue: PUV observation (8.3 m).
Dashed: shore-normal angle (264°). PUV direction converges toward shore-normal at swell frequencies, consistent with refraction.

33 / 40 instruments processed successfully (**82.5%**)

7 failures — all hardware-related:

- Instrument burial (sand accretion)
- Bent or broken mounting pipes
- Dead / depleted batteries (no usable data)

No failures attributable to the processing pipeline itself.

Site	Processed	Failed
Torrey Pines	23	3
SIO Pier	6	3
Solana Beach	5	1
Los Peñasquitos	4	0

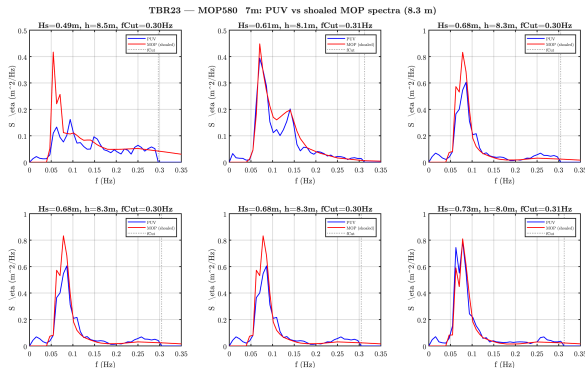
Segmentation

- **Segment length:** 2048 samples = 17.1 min at 2 Hz
- Non-overlapping segments
- **NaN policy:** segments with $>10\%$ NaN discarded; $\leq 10\%$ filled by linear interpolation
- Segment-mean pressure $\rightarrow$ water depth H
- All channels linearly detrended before spectral analysis

Typical deployment yield: TBR23 at 8.3 m depth $\rightarrow$ 5228 valid segments over 63 days (≈ 83 segments/day, 99.7% fill rate).

Spectral Estimation

- **Method:** Welch's averaged periodogram
- Sub-segments: $N_{\text{fft}} = 256$ samples (128 s)
- **Window:** Hanning, 50% overlap
- ≈ 34 effective DOF per segment
- $\Delta f \approx 0.0078$ Hz, range 0–1 Hz
- **Cross-spectra:** S_{pu} , S_{pv} for directional coefficients a_1 , b_1



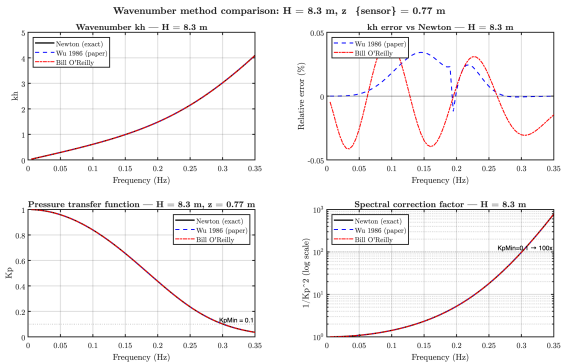
Six example segments from TBR23 MOP580 (8.3 m). Blue: PUV surface elevation spectrum $S_{\eta}(f)$ after pressure correction. Red: MOP shoaled to PUV depth. Vertical dashed lines mark f_{cut} where $K_p < 0.1$. Note the consistent excess at the swell peak (0.06–0.08 Hz).

Pressure Correction

Surface elevation spectrum from pressure:

$$S_{\eta}(f) = \frac{S_p(f)}{K_p^2(f)}, \quad K_p(f) = \frac{\cosh(k z_s)}{\cosh(k H)}$$

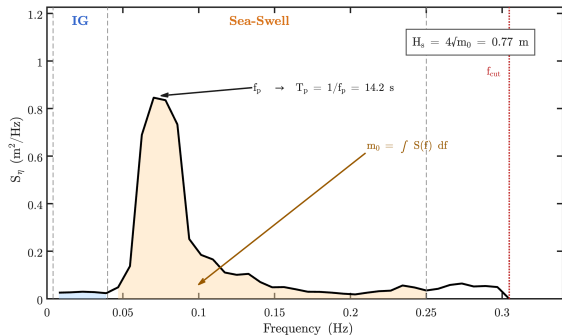
- Wavenumber $k(f)$: linear dispersion relation $\omega^2 = gk \tanh(kH)$ solved via Newton's method
- **Cutoff**: $K_p < 0.1 \Rightarrow$ set $S_{\eta}(f) = 0$ (noise suppression)
- Resulting f_{cut} recorded per segment



Wavenumber $k(f)$ at 8 m depth from three formulations: Newton's method (baseline), Wu (1986) approximation, and O'Reilly coefficients. All three produce identical results, confirming that the pressure correction is not sensitive to the wavenumber solver.

Bulk Wave Parameters

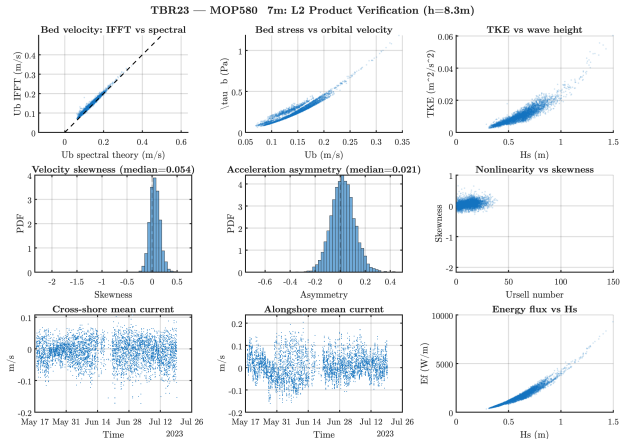
- $H_s = 4\sqrt{m_0}$, computed separately for:
 - Sea-swell: 0.04–0.25 Hz
 - Infragravity: 0.004–0.04 Hz
- Peak period $T_p = 1/f_p$ (max S_η in SS band)
- Mean period $T_{m02} = \sqrt{m_0/m_2}$
- Mean direction from a_1, b_1 (PUV cross-spectra)
- Energy flux $F = \rho g \int c_g(f) S_\eta(f) df$



Example $S_\eta(f)$ from TBR23 MOP580 (8.3 m). Dashed lines mark IG/sea-swell band boundaries (0.04, 0.25 Hz). Shaded area: m_0 integral. Arrows: f_p ($T_p = 14.2$ s). Red dotted: f_{cut} where $K_p < 0.1$. $H_s = 4\sqrt{m_0} = 0.77$ m.

Additional L2 Products

- **Near-bed velocity:** IFFT method — scale velocity FFT by $\cosh(kH)/\cosh[k(H - z_s)]$
- **Bed shear stress:** Swart (1974) friction factor, $k_s = 2.5 D_{50}$
- **Reynolds stress:** $\overline{u'w'}$, $\overline{v'w'}$; turbulent KE
- **Velocity moments:** skewness S_k , asymmetry A_s (Elgar 1985)



L2 product verification for TBR23 MOP580 (8.3 m). Top row: bed velocity IFFT vs. spectral theory (1:1), bed stress vs. orbital velocity, TKE vs. H_s . Middle: skewness and asymmetry distributions. Bottom: cross-shore and alongshore mean currents, energy flux vs. H_s .

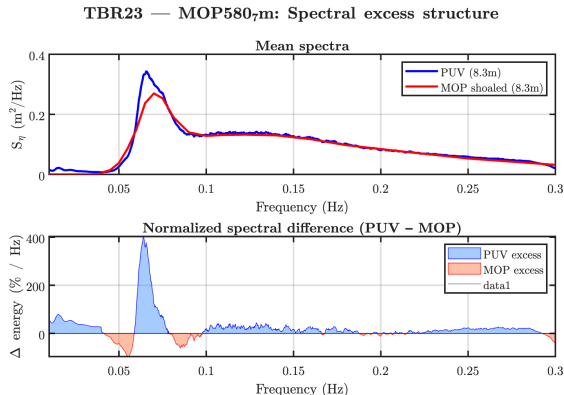
Shore-Normal Rotation

- Shore-normal angle retrieved from CDIP THREDDDS for each MOP station
- Velocities rotated: $+x$ onshore, $+y$ alongshore north
- **Fallback:** if THREDDDS unavailable, retain geographic buoy coordinates ($+x$ W, $+y$ N)
- Shore-normal frame required for cross-shore energy flux and transport calculations

Example: MOP580 shore-normal = 264° (facing WSW). After rotation, positive u = onshore (toward 084°), positive v = alongshore toward north.

MOP Wave Model

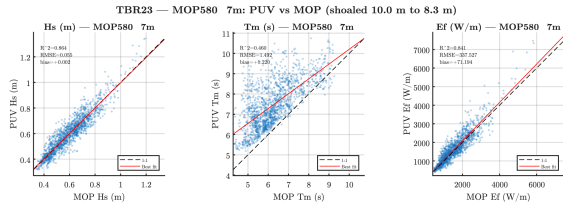
- CDIP Monitoring and Prediction (MOP) spectral wave model
- Initialized by offshore CDIP buoy observations
- Provides $S_\eta(f)$ at 10 m depth, hourly
- **Shoaling to PUV depth:**
$$S_\eta(f, h_{\text{PUV}}) = S_\eta(f, 10 \text{ m}) \times \frac{c_g(f, 10 \text{ m})}{c_g(f, h_{\text{PUV}})}$$
- PUV segments (17 min) interpolated to MOP hourly time base for comparison



Top: time-averaged PUV spectrum (blue) vs. MOP shoaled to 8.3 m (red) for TBR23 MOP580. Bottom: spectral difference after normalizing each spectrum to percent of total energy. Blue shading marks frequencies where PUV carries a larger share of energy; red where MOP does.

Bulk Comparison Results

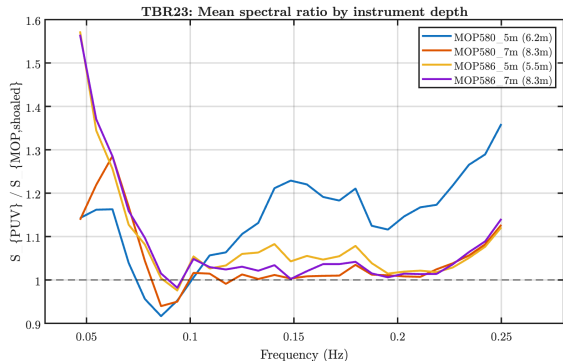
- H_s : $R^2 = 0.83\text{--}0.86$
- $\text{RMSE} < 0.06\text{ m}$
- Small positive bias ($\text{PUV} > \text{MOP}$)
- Kelp-fouled instrument: $R^2 = 0.66$, bias = $+0.037\text{ m}$



H_s scatter plot, PUV vs. MOP (shoaled) for TBR23 MOP580 at 8.3 m. Each point is one hourly comparison. Points fall slightly above the 1:1 line, indicating a systematic positive PUV bias of $\sim 3\text{--}5\%$.

Frequency-Dependent Bias

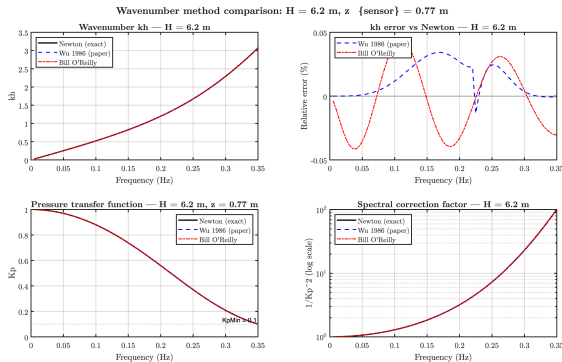
- Excess energy concentrated at **swell peak** (0.04–0.08 Hz)
- $K_p > 0.9$ at these frequencies — pressure correction is minimal
- Mid-swell band: near-unity ratio (0.99–1.03)
- **NOT** a high-frequency pressure correction artifact



Mean spectral ratio (PUV/MOP) vs. frequency for all TBR23 instruments. Ratio exceeds 1.0 at the swell peak and is near unity at higher frequencies. $K_p > 0.9$ at peak frequencies, ruling out pressure correction as the source.

Pressure Correction Investigation

- Compared three wavenumber formulations:
 - Newton's method (baseline)
 - Wu (1986) approximation
 - O'Reilly coefficients
- All produce **identical** K_p
- Pressure correction is **not** the source of the observed bias



Same comparison as previous slide but for 6 m depth, where K_p attenuation is stronger. Three wavenumber formulations remain identical. Even at shallower depth, the solver choice does not introduce bias.

Overview: Four Hypotheses

What causes the systematic positive H_s bias ($PUV > MOP$)?

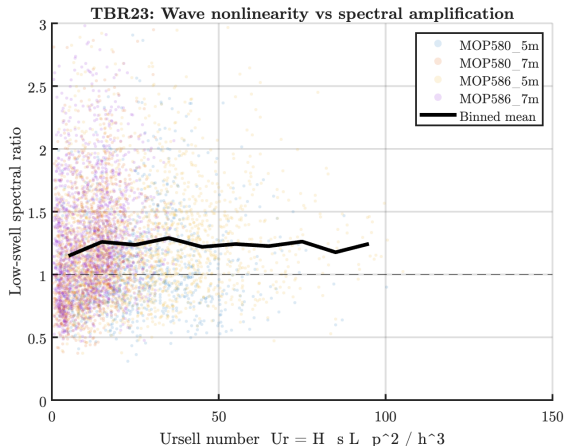
- ① **Nonlinear shoaling** — triad interactions (Ursell number)
- ② **Directional narrowing** — refraction-driven focusing
- ③ **Bound long waves** — group-forced IG contamination
- ④ **MOP spectral peak broadening** — numerical diffusion in the wave model (later eliminated: the test for it was a resolution artifact)

Each hypothesis predicts a specific correlation between a physical parameter and the observed H_s amplification ratio (PUV/MOP).

H1: Nonlinear Shoaling

- **Prediction:** nonlinear triad interactions during shoaling transfer energy to harmonics, increasing H_s beyond linear theory
- **Test:** correlate spectral ratio with Ursell number $Ur = H_s L_p^2 / h^3$
- $N = 5228$ segment-hours
- **Result:** $R^2 = 0.004$
- Median Ur : 12 at 8.3 m, 30 at 5.5 m

⇒ RULED OUT



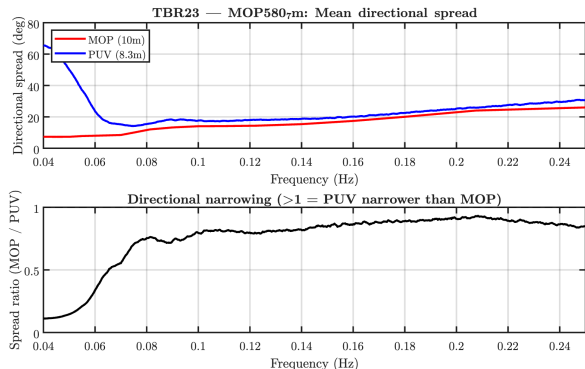
Spectral ratio (PUV/MOP in swell band) vs. Ursell number for 5228 hourly segments. No correlation ($R^2 = 0.004$). If nonlinear shoaling drove the bias, higher Ur would predict larger ratios.

H2: Directional Narrowing

- **Prediction:** refraction narrows directional spread during shoaling $\Rightarrow$ energy focusing $\Rightarrow$ higher H_s
- **Result:** PUV spread is systematically *wider* than MOP at swell frequencies (15° vs. 10°)
- Opposite to prediction
- Correlation with H_s amplification: $R^2 < 0.001$

$\Rightarrow$ RULED OUT

Wider PUV spread may reflect scattered energy at shallow depth not captured by the forward-propagating MOP model. MOP directional spread is expected to have a lower limit set by its angular resolution and propagation scheme — confirming this floor is an open item.

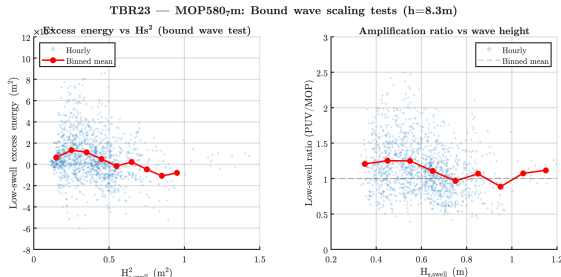


Top: mean directional spread vs. frequency. Red: MOP (10 m), Blue: PUV (8.3 m). PUV spread is wider at all swell frequencies. Bottom: spread ratio (MOP/PUV); values >1 mean PUV is wider. If directional narrowing drove the bias, PUV would be *narrower* (ratio <1).

H3: Bound Long Waves

- **Prediction:** bound wave energy $\propto H_s^2$ (Longuet-Higgins & Stewart 1962) leaks into the swell band
- **Result:** no positive correlation with H_s^2 ($R = -0.14$)
- Excess in *wrong frequency band* — at swell peak (0.04–0.08 Hz), not at group frequencies (0.01–0.03 Hz)

⇒ RULED OUT



Swell-band excess energy (PUV – MOP) vs. H_s^2 . If bound long waves were responsible, excess energy would scale positively with H_s^2 . Instead, correlation is weakly negative ($R = -0.14$). The excess also appears at swell frequencies, not IG frequencies where bound waves reside.

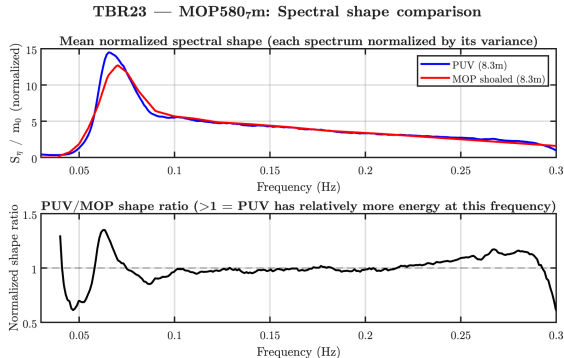
H4: MOP Spectral Peak Broadening — **RETRACTED**

- Reported: Q_p (Goda peakedness) **24%** higher in PUV than MOP
- ⇒ **ARTIFACT** — the metric interpolated MOP's ~20 coarse bins *up* onto the PUV's 3601-point grid
- A PUV spectrum compared against a degraded copy of **itself** through that path yields Q_p ratio **1.225** — larger than the whole reported effect
 - On a resolution-matched grid, across 62 of 65 records: Q_p ratio **1.008** (IQR 0.980–1.039), $p = 0.44$ against 1

There is no peak-shape bias. The legacy path reproduces the 1.225 self-comparison artifact across the whole catalog (1.203, IQR 1.161–1.238).

$Q_p = \frac{2}{m_0^2} \int f S_\eta^2 df$ (Goda, 1970). Retracted 2026-07-25; see findings_resolution_artifact_2026-07-24.md. Replacement code:

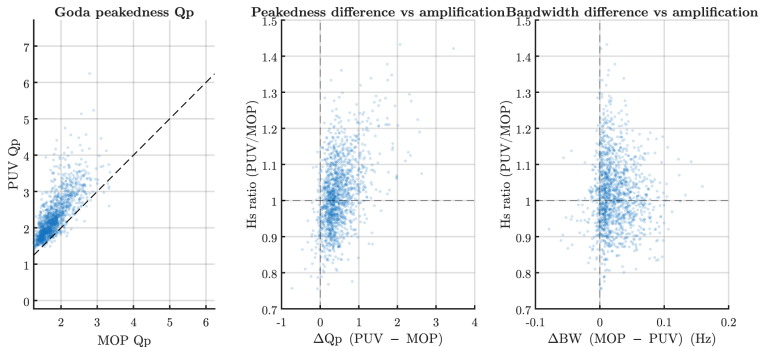
validation/compare_shape_matched.m



Frequency-normalized spectral shape (centered on f_p , normalized by peak density). Blue: PUV. Red: MOP. The PUV peak is taller and narrower; MOP spreads energy laterally. This shape difference accounts for the integrated H_s bias.

Peakedness Diagnostic — **RETRACTED**

TBR23 — MOP580_{7m}: Spectral peakedness analysis



This figure is computed on the interpolated grid and shows the artifact, not a physical effect. It was read as: hours when the PUV spectrum is more peaked relative to MOP correspond to hours with larger positive H_s bias ($R = 0.44$, TBR23 MOP580, 8.3 m). Because both axes inherit the same resolution mismatch, the correlation does not establish a shape bias. Regenerate on a matched grid with `validation/compare_shape_matched.m` before using.

Mechanism attribution

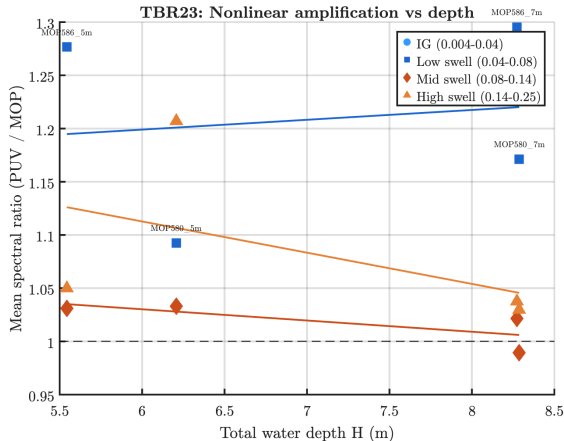
Mechanism	Evidence	Status
Nonlinear shoaling	$\rho(Ur, \nu) = +0.332$; ratio peaks 1.34 at $f/f_p = 2.1$ ($H_s/h > 0.12$)	the mechanism
Bound long waves	$\rho = +0.913$ vs. H_s/h per record; partial $+0.149$ after H_s/h	same triads
Directional spread	σ_1 ratio 1.156 ($p = 2.6 \times 10^{-6}$), uncorrelated with the discrepancy	separate bias
Peak broadening	matched-grid Q_p ratio 1.008	eliminated

72,948 hours, 61 records. The spectral-width ratio is 1.000 below $H_s/h = 0.04$ and 1.100 above 0.20: the model tracks the observations while the waves are linear.

Cross-Deployment: what survives the matched-grid test

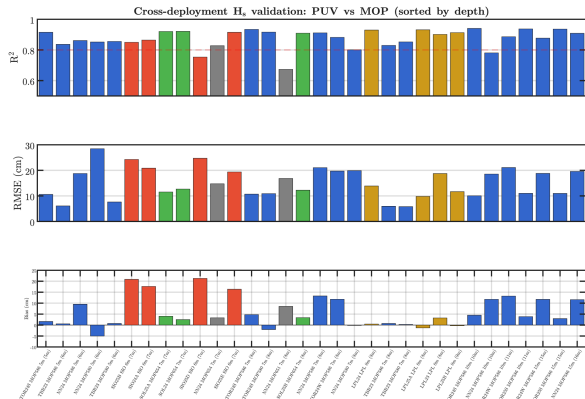
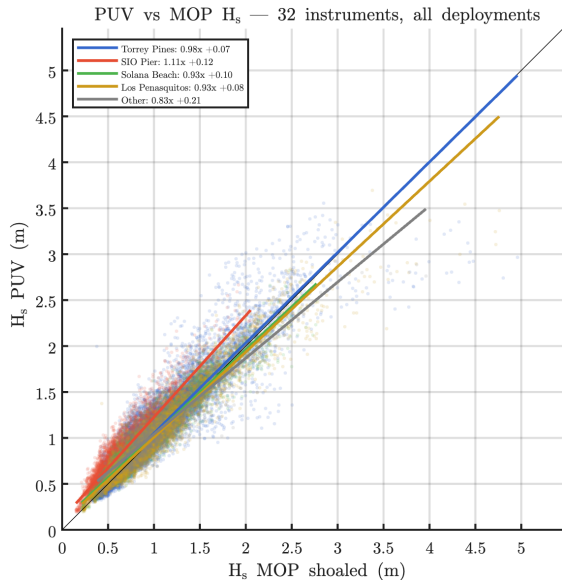
- **Not peak broadening.** Reproducing an artifact across 11 instruments reproduces the artifact, not a mechanism
- Matched grid, 62 of 65 records: Q_p ratio 1.008, $p = 0.44$ — **no peak-shape bias**
- **What is real:** spectral width ν ratio 1.060 ($p = 4.6 \times 10^{-9}$) and peak density 1.064 ($p = 8.6 \times 10^{-11}$)
- PUV carries a slightly sharper peak *and* more high-frequency energy; MOP holds more in the near-peak shoulders. Q_p integrates over the band and cannot see that redistribution
- **Depth trend survives a band control:** $\rho(\nu, h) = -0.42$ ($p = 0.0008$) at a fixed 0.18 Hz upper limit, below every f_{cut} in the catalog

Records within a site are not independent — treat p as indicative.



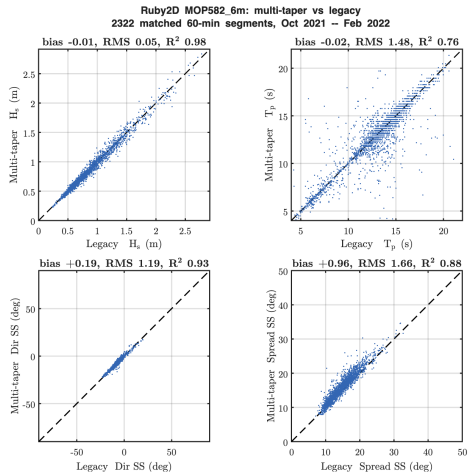
Spectral ratio (PUV/MOP) by frequency band vs. total water depth for TBR23 instruments. Low-swell band (0.04–0.08 Hz, triangles) shows the largest amplification (>1.1), decreasing weakly with depth. Mid-swell and sea bands cluster near 1.0.

Bulk H_s Validation: All Deployments



Per-instrument R^2 , RMSE, and bias sorted by depth. Positive bias (PUV > MOP) is consistent across instruments. RMSE is generally < 6 cm.

Pipeline Cross-Validation: Ruby2D Head-to-Head



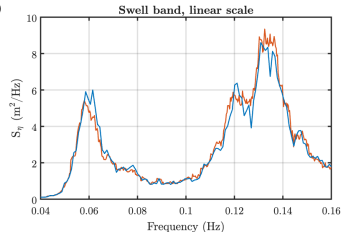
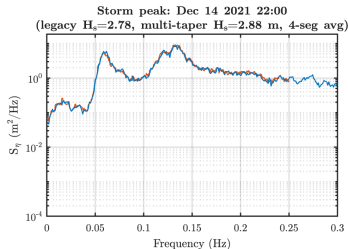
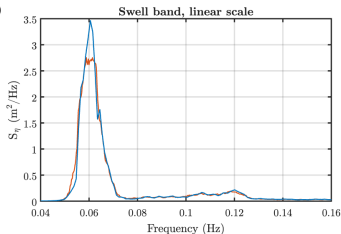
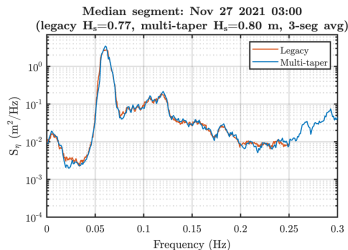
- Validates the new pipeline against an **independent prior implementation**
- H_s : RMS **5 cm**, $R^2 = 0.98$, bias -9 mm
- Direction: RMS **1.2°** , $R^2 = 0.93$
- Spread: RMS 1.7° , $R^2 = 0.88$ (new pipeline 1° wider)
- T_p : RMS 1.5 s, $R^2 = 0.76$ (after band-limit filter)

Bottom line: the multi-taper pipeline reproduces a working prior pipeline within instrument noise. Prior Ruby2D bulk results stand.

Ruby2D MOP582_6m, Oct 2021 – Feb 2022. 2,322 matched 60-min segments, the new pipeline (vertical) vs. the legacy archived L2 (horizontal). Dashed line is 1:1.

Ruby2D Spectral Shape Comparison

Ruby2D MOP582_6m: representative spectra



- Two representative segments:
 - Median, $H_s = 0.77$ m
 - Storm peak, $H_s = 2.78$ m (bimodal seas)
- Integrated H_s from spectra agrees within **4%** in both cases
- Legacy spectra (red) carry single-periodogram chi-square noise across the band
- Multi-taper (blue) is smoother — it averages over $K = 7$ DPSS tapers, $\sim 7\times$ variance reduction per bin
- **Same physical spectrum, lower variance estimate**

L3 Architecture

L3 transforms L2 spectra into forcing metrics relevant to sediment transport. Four sub-modules, each deployment-agnostic:

L3a: Band Decomposition

- IG: 0.004–0.04 Hz
- Swell: 0.04–0.10 Hz
- Sea: 0.10–0.25 Hz
- E_f and H_s per band, per segment

L3b: Storm Detection

- Threshold exceedance + duration
- MOP gap-filling for instrument failures
- Event metrics: peak H_s , duration, cumulative F

L3c: Transport Proxies

- Bottom energy flux F_b
- Shields parameter θ_s
- Rouse number R_o
- Mobilization fraction (% time $> \theta_{cr}$)

L3d: Current Decomposition

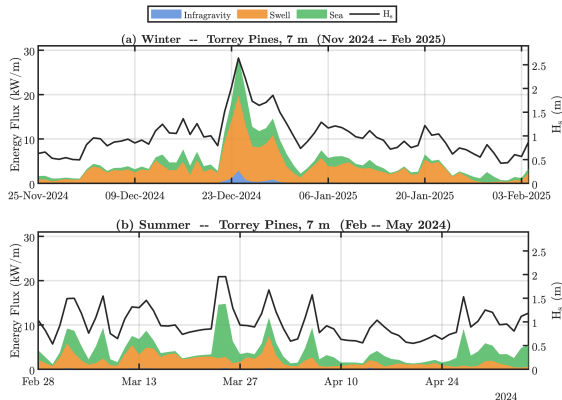
- Tidal harmonics (τ_{tide})
- Subtidal (wave-driven) residual
- Undertow + longshore current

Energy conservation verified: $E_{f,\text{total}}/E_{f,\text{L2}} = 1.0000$ for all 33 instruments (no energy created or lost in band partitioning).

Seasonal Energy Partitioning

- **Winter:** swell-dominated (65–90% of energy flux)
 - North Pacific swell, $T_p > 12$ s
 - H_s routinely > 1.5 m at 8 m depth
- **Summer:** sea-dominated or equal ($\sim 50/50$)
 - Southern-hemisphere swell + local sea breeze
 - H_s typically 0.4–0.8 m
- **Spring:** transitional
- Consistent across all 5 sites

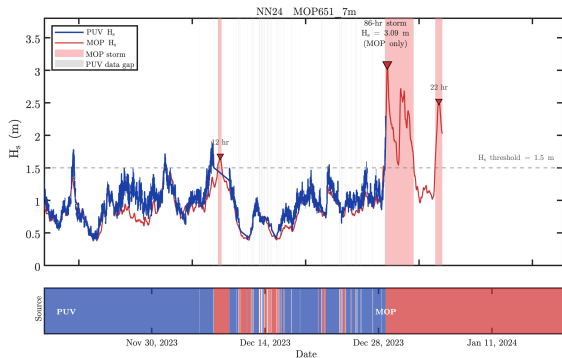
Band	Frequency	Period
IG	0.004–0.04 Hz	25–250 s
Swell	0.04–0.10 Hz	10–25 s
Sea	0.10–0.25 Hz	4–10 s



Daily-averaged energy flux (kW/m) at Torrey Pines MOP586, 7 m depth. (a) Winter (TOR24W): swell-dominated, peaks to 30 kW/m. (b) Summer (TOR24S): lower energy, higher sea-band fraction. Black line: H_s (right axis).

Storm Detection + MOP Gap-Filling

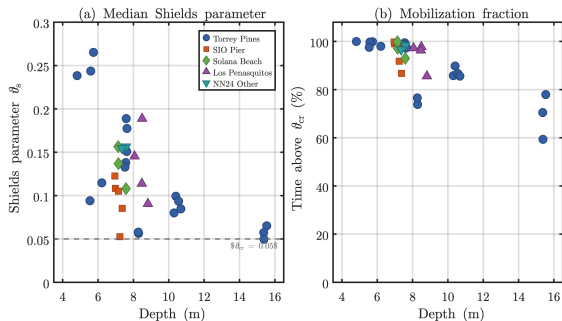
- **Example:** TOR23W/SOL23 Dec 28, 2023 storm
 - H_s up to 3.6 m, 74-hour duration
 - Caused instrument failures at 2 sites
- MOP fills data gaps from instrument damage during storms
- MOP651_7m: detected 0 PUV storms, MOP found 3 (86-hour event)
- Gap-filling uses the validated MOP spectra with the known small bias



SOL23 MOP651 (7 m). Blue: PUV H_s . Red: MOP H_s . Gray bands: PUV data gaps. Pink bands: MOP-detected storms (all 3 missed by PUV). Largest event: 86 hr, peak $H_s = 3.09$ m. Bottom bar: data source (PUV vs. MOP).

Transport Proxy Depth Dependence

- Shields parameter decreases offshore:
 - $\theta_s = 0.24$ at 5.6 m
 - $\theta_s = 0.065$ at 15.5 m
- Mobilization: 100% at 5 m, 78% at 15 m (winter)
- Rouse number: bedload-dominated across all sites ($R_o > 2.5$)
- Cumulative F_b : winter $3\times$ summer at equivalent depth

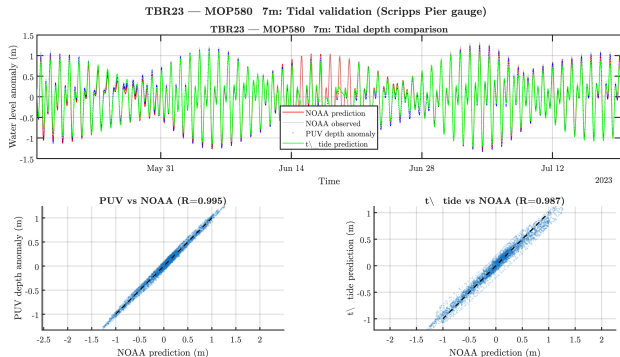


(a) Median Shields parameter vs. depth for all 33 instruments, colored by site. Dashed line: critical Shields ($\theta_{cr} = 0.05$).

(b) Mobilization fraction (% of segments exceeding θ_{cr}) vs. depth.

Tidal Validation

- NOAA Scripps Pier tide gauge vs. PUV depth: $R = 0.995$
- Confirms PUV clocks are UTC (lag = 0.0 hours)
- Subtidal cross-shore current: consistently offshore (undertow)
- Tidal currents small (<0.05 m/s) vs. wave-driven flows (~ 0.10 – 0.30 m/s)



Top: PUV depth anomaly (blue) vs. NOAA Scripps Pier tide gauge (green) and t_tide prediction (black). Bottom left: PUV vs. NOAA scatter ($R = 0.995$). Bottom right: t_tide prediction vs. NOAA ($R = 0.987$). Zero lag confirms PUV timestamps are UTC.

L4: Infragravity and Bound Wave Dynamics

Objective: Separate forced (bound) and free infragravity waves using the co-located pressure and velocity measurements. **Approach:**

- Bispectral analysis — bicoherence at swell-IG coupling frequencies identifies energy transfer from swell triads to the IG band
- Phase relationship between IG pressure and wave group envelope distinguishes bound (locked phase) from free (random phase) IG waves
- IG energy flux and direction from S_{pu} and S_{pv} cross-spectra in the IG band
- Reflection coefficient from shoreward/seaward energy flux ratio

Why it matters:

- IG waves dominate inner-surfzone runup and overwash
- Bound IG energy transfers to free waves during breaking — the ratio controls IG amplitude at the shoreline
- MOP does not model IG waves; PUV measures them directly
- Multi-instrument arrays (TOR23W, TOR24) enable cross-shore IG coherence analysis

Reference: Herbers et al. (1995); implementation informed by Athina Lange's Ruby2D code.

L5: PUV–Altimeter Integration

Objective: Correlate wave forcing (L2/L3 products) with bed-level change from co-located acoustic altimeters. **Data sources:**

- Aquatec AA400 altimeters (acoustic bed tracking)
- Simrad EA400 echologgers
- Co-located with PUV instruments at multiple Torrey Pines and Solana Beach sites

Analysis plan:

- 1 Time-align altimeter bed level with L2/L3 wave forcing segments
- 2 Test 8 transport relationships: F_b , θ_s , Shields exceedance, H_s , $S_k \cdot U_b^3$, storm cumulative flux, tidal phase, event-scale response
- 3 Estimate transport rates from bed-level change + continuity equation

Key question: Which wave forcing metric best predicts observed bed-level change?

- Is it total energy (F_b)?
- Is it the nonlinear shape ($S_k \cdot U_b^3$)?
- Does it depend on depth or grain size?

Feedback to L2: Time-varying bed level from altimeter corrects the pressure sensor offset (doffp), improving K_p accuracy as sediment accretes or erodes around the instrument. Starting point: TOR24S MOP586

(co-located, multi-depth array). See [docs/altimeter_correlation_plan.md](#).

TBR23 Context: Post-Storm Recovery

Background:

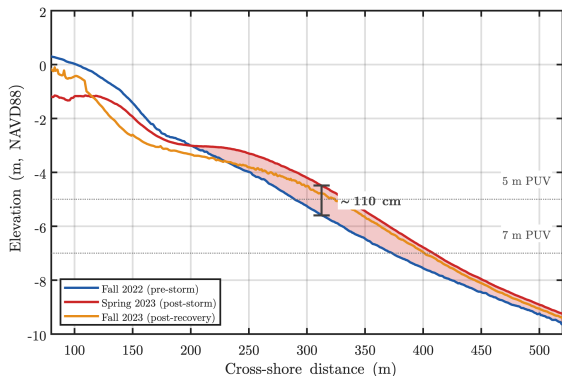
- Winter 2022–2023: sequence of atmospheric rivers deposited ~ 85 cm of sand at 5 m depth (Torrey Pines)
- Largest winter accretion in 25-year MOP survey record
- TBR23 deployment (May–Aug 2023) captures the subsequent summer recovery phase

Field campaign:

- 22 jetski bathymetric surveys (every 3 days)
- 4 PUV instruments: 2 at 5 m, 2 at 7 m depth
- Co-located with MOPS truck lidar surveys (weekly)

Central question:

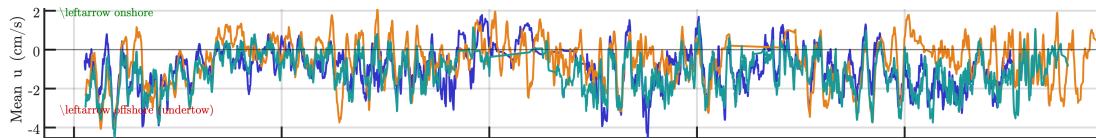
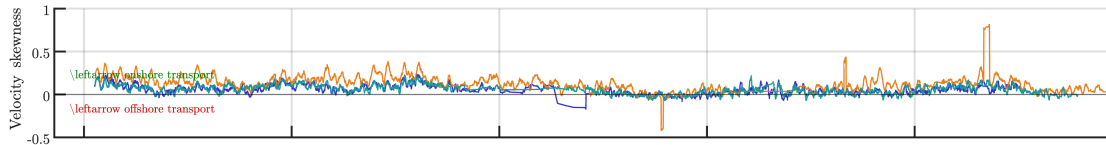
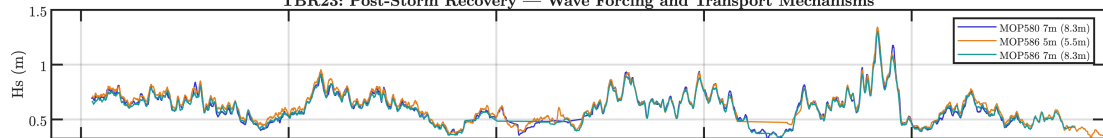
What wave processes drive post-storm bathymetric recovery, and why does recovery only occur at certain depths?



Alongshore-averaged profiles at Torrey Pines (MOP 580–586).

Wave Forcing and Bed Response

TBR23: Post-Storm Recovery — Wave Forcing and Transport Mechanisms



Competing Transport Mechanisms

Two processes act simultaneously:

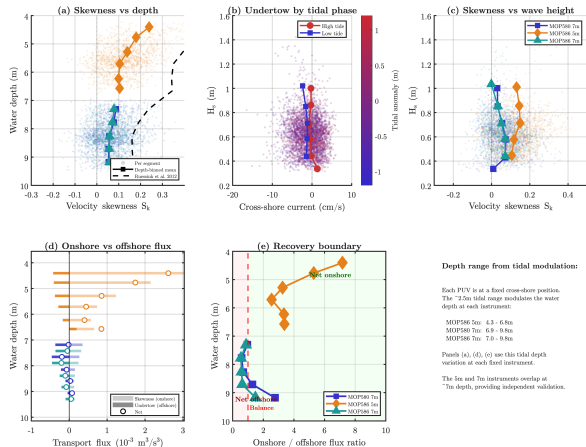
Onshore: velocity skewness ($S_k > 0$)

- Peaked wave crests produce stronger onshore orbital velocities than troughs
- Bedload transport scales as U_b^3 weighted by skewness
- Drives recovery (shoreward sand movement)

Offshore: undertow

- Wave-driven mean return flow below trough level
- Opposes recovery
- Strengthens with increasing H_s

TBR23: Competing Transport During Post-Storm Recovery

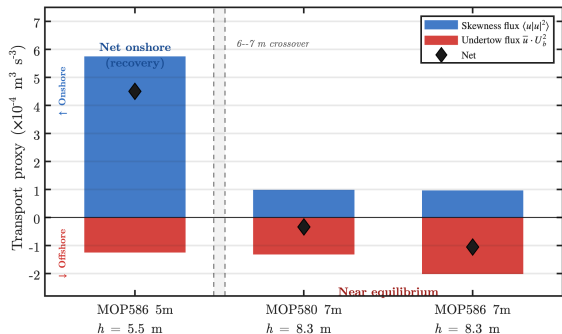


Cross-shore transport decomposition for TBR23 instruments.

Skewness-driven flux (blue, onshore) vs. undertow-driven flux (red, offshore). At 5.5 m depth, skewness dominates; at 8.3 m, the two are

Recovery Boundary at 6–7 m Depth

- At 5.5 m: skewness flux 3–4× stronger than undertow → net onshore (recovery)
- At 8.3 m: undertow slightly dominates → near-equilibrium
- Crossover at ~6–7 m depth
- Explains: mid-July recovery pulse at 5 m site but not at 7 m sites
- Tidal modulation provides depth range at each fixed instrument

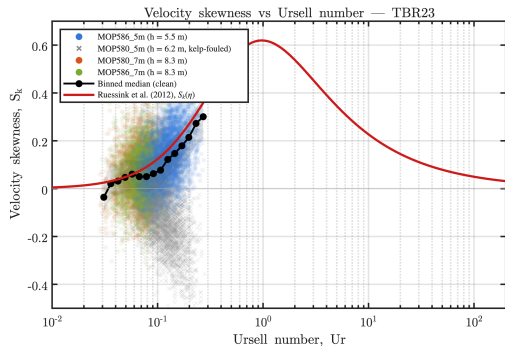


Transport proxy decomposition for TBR23. Blue: skewness flux $\langle u|u|^2 \rangle$ (onshore). Red: undertow flux $\bar{u} \cdot U_b^2$ (offshore).

Diamonds: net. At 5.5 m skewness dominates 4:1; at 8.3 m the two balance. Dashed lines: inferred 6–7 m crossover.

Key Physics

- Skewness follows Ruessink et al. (2012) depth parameterization
 - S_k increases with decreasing kh
 - Observed values match predicted curve
- Undertow strengthens at high tide (more wave energy reaches instrument depth)
- Low tide reduces undertow, enabling onshore recovery
- MOP580_5m excluded: kelp fouling produced artifact negative S_k



Velocity skewness vs. Ursell number for all TBR23 segments. Binned medians (black dots) from 3 clean instruments track the Ruessink et al. (2012) curve (red, surface-elevation S_k). Gray: kelp-fouled MOP580_5m with anomalous negative skewness.

Key Findings

- ① **Pipeline validated:** 33/40 instruments processed (82.5%), strong agreement with MOP ($R^2 > 0.83$, $RMSE < 0.06\text{ m}$)
- ② **Bias identified:** small systematic positive H_s bias concentrated at the swell spectral peak, *not* from the pressure correction
- ③ **Mechanism: nonlinear shoaling (Ursell)**, on the elimination redone 2026-07-27 over 72,948 hours / 61 records. The earlier “MOP peak broadening” answer was a resolution artifact of our own metric, retracted 2026-07-25
- ④ **Cross-deployment:** matched-grid Q_p ratio 1.008 ($p = 0.44$) across 62 of 65 records — no peak-shape bias. Spectral width and peak density each run $\sim 6\%$ high, and the width discrepancy deepens toward shore ($\rho = -0.42$, band-controlled)
- ⑤ **Full L1–L3 pipeline operational:** all 33 instruments processed through forcing characterization, storm detection, and transport proxy computation
- ⑥ **TBR23 transport balance:** competing skewness vs. undertow, recovery boundary at 6–7 m depth

Implications for TBR23 Paper

- Pipeline is validated and ready for production use
- Bias is understood and attributable to the model, not the measurement — no correction needed for PUV data
- Spectral shape finding *supports* using PUV as ground truth for nearshore wave characterization
- L2 products (bed velocity, stress, moments) are ready for sediment transport analysis
- L3 forcing characterization provides storm context, seasonal partitioning, and transport proxies
- Next: quantitative transport model (Bailard / Hoefel & Elgar) to compare computed fluxes against observed bed change

Immediate (TBR23 Paper):

- Implement Bailard / Hoefel & Elgar transport model using L2 products
- Compare computed transport rates against survey-to-survey bed change
- Draft results and discussion sections
- Process grain size data (laser particle analyzer) for site-specific D_{50}

Pipeline extensions:

- **L4:** Bispectral analysis for bound/free IG wave separation
- **L5:** PUV–altimeter integration (8 transport relationships to test)
- **Wave dynamics paper:** nonlinear shoaling as the bias mechanism, the surviving spectral-width and peak-density excess, SIO canyon focusing, depth trends

Questions?

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GitHub: [holdenlesliebole/Nortek_Vector_PUV_Pipeline](https://github.com/holdenlesliebole/Nortek_Vector_PUV_Pipeline)

All data, code, and figures reproducible from the repository.